

CHI SQUARE FOR Goodness of fit

In 2010 census of the city, the weight of the individuals in a small city were found to be the following

< 50 kg	50-75	> 75
20%	30%	50%

In 2020, weight of $n = 500$ individuals were sampled. Below are the results

< 50	50-75	> 75
140	160	200

Using $\alpha = 0.05$, would you conclude the population differences of weights changed in the last 10 years?

Ans $\rightarrow$

2010
Expected

< 50	50-75	> 75
140	160	200

$n=500$
8020
observed

< 50	$50 - 75$	> 75
140	160	200

< 50	$50 - 75$	> 75
0.2×500 $= 100$	0.3×500 $= 150$	0.5×500 $= 250$

① Null Hypothesis: H_0 : The data meets the expectation

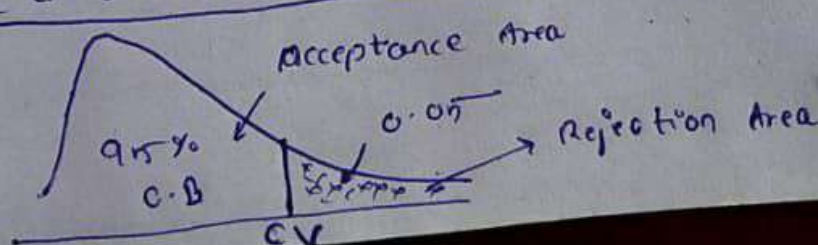
Alternate Hypothesis: H_1 : The data does not meet the expectation

② $\alpha = 0.05$ C.D. = 95%

③ Degree of freedom

$$df = K - 1 = 3 - 1 = 2$$

④ Decision Boundary



CV $\rightarrow$ Critical value
5.991
CV $\rightarrow$ critical value

$$CV = 5.991$$

If χ^2 is greater than 5.99,
Reject H_0 else

we fail to reject the Null
Hypothesis.

⑤ Calculate Chi Square Test Statistics

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(140 - 100)^2}{100} + \frac{(160 - 150)^2}{150} + \frac{(200 - 250)^2}{250}$$

$$= \frac{1600}{100} + \frac{100}{150} + \frac{2500}{250}$$

$$= 16 + 0.66 + 10$$

$$= 26.66$$

$$\chi^2 = 26.66$$

If $\chi^2_{H_0}$ is greater than 5.99, Reject
else

we fail to reject the null hypothesis

$26.66 > 5.99$, Reject H_0

Answered

The weight of 2020 population are different than those expected in the 2010 population.